

Andrew W. Shenstone

Shenstone, Andrew - Application for Product (Front-End) Engineer @ Orbis

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SUMMARY

Senior software engineer with strong front-end depth across React, TypeScript, Next.js, and Tailwind, plus full-stack context from Bloom.wine, Tilia, and founder-led Plantfolio. I focus on clear UX for complex flows, craft, and close work with product and design.

I am applying through Orbis for a front-end-leaning product engineer role: early-stage ownership of core UI, interactive surfaces, and collaboration with founders. That lines up with how I have shipped before—Plantfolio from zero and design-system adoption on larger teams.

SKILLS

Frontend: React, TypeScript, Next.js (App Router, RSC), HTML/CSS, responsive layouts, Tailwind CSS and Shadcn UI; Chakra UI adoption at Tilia; component libraries and design-system habits with Figma-informed workflows, Complex UIs: React Hook Form, Zustand, Framer Motion; performance-conscious data fetching

Product & delivery: Founder-led Plantfolio: discovery through production, iteration with users, AI-powered features in approachable UX, Cross-functional work with product and design; scope trade-offs and measurable outcomes, Testing and quality (Vitest, Cypress); code review and pairing

Backend & real-time context: Node.js, REST and GraphQL APIs (Bloom, Tilia); integrations and webhooks, Async and event-driven patterns (Inngest, background jobs); WebSockets/SSE called out in postings are not my deepest specialty—I implement what the stack requires with clear contracts

AI/ML: OpenAI API, AI SDK, Vector Embeddings, Natural Language Processing, Retrieval Augmented Generation (RAG)

EDUCATION

B.S. Finance, Santa Clara University — 2010

EXPERIENCE

Founder & Lead Engineer

Plantfolio | November 2023 - Present

Product Leadership & Architecture

- Architected and built a comprehensive plant management platform from the ground up, serving as both technical lead and product owner
- Owned the full feature cycle: joined customer calls to identify friction points, architected solutions, implemented, tested, and verified success in production
- Designed and implemented a sophisticated monorepo architecture with 12+ packages, enabling rapid development and maintainable codebases
- Led product strategy and user experience decisions, making product micro-decisions and navigating ambiguity to deliver value iteratively

Technical Achievements

- **AI Integration:** Built advanced AI-powered features including plant identification from photos, contextual chatbot (Sage), and smart text extraction from plant tags using OpenAI's vision and language models
- **Modern Frontend Architecture:** Implemented Next.js with React, React Server Components, and TypeScript, achieving exceptional performance and developer experience
- **Complex Data Modeling:** Designed and implemented sophisticated PostgreSQL schemas with Prisma, handling complex relationships between users, plants, folios, and media with 15+ interconnected models

- Supabase to Neon: Shipped on Supabase (PostgreSQL with Row-Level Security policies for multi-tenant isolation) before migrating the data layer to Neon; strong hands-on experience with RLS and Postgres-centric access control
- Content Management: Created sophisticated content management system with rich text editing, plant tagging, and flexible data structures

Scale & Performance

- Managed a database of 40,000+ plant species with advanced search capabilities using Algolia and vector embeddings
- Implemented comprehensive media management system with AWS S3, supporting image optimization, galleries, and metadata extraction
- Built robust authentication and authorization systems with Clerk, supporting multiple user types and subscription tiers
- Optimized application performance through React Server Components, efficient data fetching, and strategic caching

Client Case Study - AI Signage

- Delivered a production rollout for an independent garden center to automate plant signage generation; one independent retail customer in ongoing production who has been an enthusiastic reference
- Enabled bulk plant import (plain text) to printable signs in seconds; reduced a weekend-long manual process to minutes after initial setup
- Demonstrated ability to translate customer requirements into scalable technical solutions and shorten feedback loops

Full-Stack Ownership

- Developed complete payment integration with Stripe, subscription management, and user onboarding flows
- Built comprehensive email system with React Email, supporting transactional emails, notifications, and marketing campaigns
- Implemented advanced search and filtering capabilities with Algolia, supporting complex plant queries and user preferences

Code Quality & Leadership

- Established coding standards and architectural patterns that enabled rapid, consistent development
 - Implemented comprehensive testing strategy with Vitest and Cypress, maintaining high code quality
 - Built reusable component library and design system, ensuring consistent UI/UX across the platform
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Senior Software Engineer

Tilia | January 2023 - August 2023

- Partnered with cross-functional teams in product and design to revamp critical user-facing features
 - Built and shipped work in support of KYC, identity verification, and risk-adjacent tooling as part of Tilia's payments platform (tools team context)
 - Introduced and evolved a GraphQL API on the platform while consuming it from our product—hands-on with both server-side schema/resolvers and client integration
 - Spearheaded and guided a team through the successful adoption of Chakra UI, reducing bespoke components and establishing cohesive design patterns
 - Mentored engineers through pairing and code review in an informal technical lead capacity (no direct reports)
 - Maintained exceptional unit test coverage of over 90% for the Tools team application, alongside comprehensive end-to-end testing
 - Led the swift creation and deployment of a homepage reporting dashboard within the initial two months of joining
 - Role ended in a company-wide layoff affecting approximately one-third of employees
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Senior Software Engineer

- Architected a greenfield Next.js application for the primary admin interface of the Bloom product; much of that stack remains in production across many winery clients today
 - Engineered Ruby on Rails GraphQL API endpoints shared by multiple applications across mobile and web devices
 - CI/CD with CircleCI for build, test, and deployment workflows alongside the Next.js and Rails stack
 - Mentored engineers through pairing and code review in an informal technical lead capacity (no direct reports)
 - Collaborated with designers using Figma, establishing component library standards usable by both engineering and design
 - Integrated DatoCMS and extended application functionality through third-party API integrations (Shopify, Mailchimp, Stripe, Shipstation, ActiveCampaign); all Bloom winery clients were on Shopify Plus
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Software Engineer II

Tune Inc - Marketplace Automation | August 2019 - May 2020

- Built and operated ETL-style data pipelines and integrations supporting marketplace automation workflows
 - Automated a client workflow using a Node.js/Express API and React.js, reducing the time to complete the task from 10-15 minutes to 10-30 seconds
 - Modernized the primary legacy React.js application by moving from a custom Webpack process to react-scripts, resulting in a 50% reduction in production build size and a 60% reduction in NPM dependencies
 - Reduced development machine CPU usage from 75% to <5% by simplifying the build pipeline and dependency graph
 - Role ended in a company-wide layoff
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Software Developer

Ply Interactive Inc | June 2014 - July 2019

React Native

- Shipped multiple production React Native applications for agency clients—cross-platform mobile apps delivered end to end
- Washington State Fair: React Native app in sustained multi-year production; very high traffic during fair season, from tens of thousands of users into the hundreds of thousands cumulatively over seasons

Web & platforms

- Developed numerous web applications including Georgetownbeer.com, WashingtonWine.org, TwoWolvesWine.com, Novaerus.com, WellAir, and Plasma-air.com
- Built a standalone Ruby gem Content Management System, enabling simplified content management within Ruby on Rails projects
- Transitioned the agency to modern technologies like GraphQL, Gatsby.js, and React.js, improving delivery times for new projects
- Bloom.wine Product Development: Built extensive reporting features using complex PostgreSQL queries and interactive filtering, saving clients thousands of hours on previously manual tasks
- Engineered a flexible Wine Club membership structure for wineries, allowing for diverse client configuration and broadening the range of potential sales targets
- Communicated and enforced the standards of Shopify's Polaris React.js component library with designers, resulting in efficiencies gained from leveraging pre-built components

COMPANY REFERENCE

Plantfolio (2023-Present) - AI-powered plant management platform with 40,000+ plant database serving both consumer garden management and independent garden center services - plantfolio.garden/

Tilia (2023) - All-in-one payment platform for digital economies - tilia.io/

Bloom.wine (2020-2022) - Commerce services for the alcohol industry; all clients on Shopify Plus - bloom.wine/

Tune Inc - Marketplace Automation (2019-2020) - SaaS platform for managing marketing partnerships across mobile and web - tune.com/

Ply Interactive Inc (2014-2019) - Boutique agency: web and production React Native apps (e.g. Washington State Fair at major scale); developed Bloom.wine product internally (2017-2019)